Electrolysis of Molten Compounds

Exam code: 0620 & 0971

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Electrolysis of molten compounds

- A binary ionic compound is one consisting of just two elements joined together by ionic bonding
- When these compounds undergo electrolysis they always produce their corresponding elements
- To predict the products made at each electrode, first identify the ions
- The **positive** ion will migrate towards the **cathode** and the **negative** ion will migrate towards the **anode**
- Therefore, the **cathode** product will always be the **metal,** and the product formed at the **anode** will always be the **non-metal**

Electrolysis of molten lead(II) bromide

Method:

- Add lead(II) bromide into a beaker and heat it so it will turn molten, allowing ions to be free to move and conduct an electric charge
- Add two graphite rods as the electrodes and connect this to a power pack or battery
- Turn on the power pack or battery and allow electrolysis to take place
- Negative bromide ions move to the positive electrode (anode) and each loses one electron to form bromine molecules. There is bubbling at the anode as brown **bromine gas is given off**
- Positive lead ions move to the negative electrode (cathode) and gain electrons to form a **grey lead metal** which deposits on the surface of the electrode

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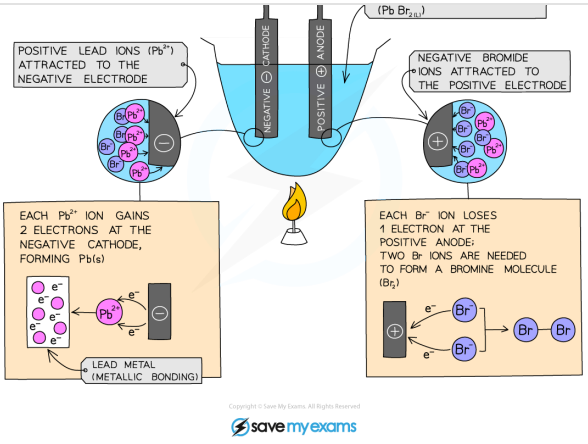

Diagram showing the electrolysis of lead(II) bromide

Worked Example

Identify the product formed at the anode and cathode during the electrolysis of molten potassium chloride.

Answer:

- The ions present are potassium (K^+) and chloride (Cl^-)
- The chloride ions are attracted to the anode and form chlorine gas
- The potassium ions are attracted to the cathode and form potassium metal

Examiner Tips and Tricks

Remember: Electrodes need to be inert such as graphite or platinum so that they don't participate in a side reaction with the electrolyte.

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